

## Exact value equations



1

a No

b Yes

2

a Zero

b Zero

c Zero

3

a  $\theta = 45^\circ$

b  $\theta = 60^\circ$

c  $\theta = 60^\circ$

d  $\theta = 60^\circ$

4

a  $\theta = 135^\circ, 225^\circ$

b  $\theta = 60^\circ, 300^\circ$

c  $\theta = 90^\circ, 270^\circ$

d  $\theta = 30^\circ, 150^\circ$

e  $\theta = 180^\circ$

f  $\theta = 225^\circ, 315^\circ$

g  $\theta = 135^\circ, 225^\circ$

h  $\theta = 240^\circ, 300^\circ$

i  $\theta = 90^\circ$

j  $\theta = 60^\circ, 240^\circ$

k  $\theta = 0^\circ, 180^\circ, 360^\circ$

l  $\theta = 150^\circ, 330^\circ$

m  $\theta = 135^\circ, 315^\circ$

n  $\theta = 60^\circ, 300^\circ$

o  $\theta = 120^\circ, 240^\circ$

p  $\theta = 45^\circ, 135^\circ$

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## Non-exact value equations

5

a  $\theta = 37^\circ$

b  $\theta = 40^\circ$

c  $\theta = 36^\circ$

d  $\theta = 21^\circ$

e  $\theta = 66^\circ$

6

a  $\theta = 25^\circ, 335^\circ$

b  $\theta = 143^\circ, 217^\circ$

c  $\theta = 220^\circ, 320^\circ$

d  $\theta = 69^\circ, 111^\circ$

e  $\theta = 35^\circ, 215^\circ$

f  $\theta = 144^\circ, 324^\circ$



## Different domains

7

a Zero

b Zero

8

a Two

b Two

9

a  $\theta = 45^\circ$

b  $\theta = -30^\circ$

c  $\theta = -45^\circ$

d  $\theta = 45^\circ$

10

a  $\theta = 146.20^\circ, -146.20^\circ$

b  $\theta = 120^\circ, -120^\circ$

c  $\theta = 60^\circ, 120^\circ$

d  $\theta = 13.77^\circ, 166.23^\circ$

e  $\theta = 45^\circ, -135^\circ$

f  $\theta = 51.05^\circ, -128.95^\circ$

11

a  $\theta = 381^\circ, 519^\circ$

b  $\theta = 426^\circ, 606^\circ$